

Lahore AQI Predictor

72-Hour Serverless Machine Learning Forecasting System

Location: Lahore, Pakistan (31.5204 N, 74.3587 E) | Stack: 100% Serverless

Live App: <https://aqi-project-c3cg2qetkj3sdwcvzcdwcf.streamlit.app/>

1. Executive Summary

Lahore is consistently ranked among the most air-polluted urban areas globally, with severe winter smog pushing the Air Quality Index (AQI) above hazardous levels (>300 to 500). This project implements an end-to-end, 100% serverless machine learning system that automates hourly telemetry ingestion, computes 35 atmospheric and lag features, trains time-series models, and delivers dynamic 72-hour recursive AQI forecasts through an interactive web dashboard and REST API.

2. System Architecture & Technology Stack

The system employs a decoupled dual-backend architecture: continuous hourly feature ingestion is managed by a high-speed serverless Supabase PostgreSQL Feature Store, while model artifact versioning and metadata are tracked via Hopsworks Model Registry.

Technology	Role	Purpose & Functionality
Scikit-Learn & XGBoost	Model Suite	Ridge, Random Forest & Gradient Boosted time-series models
Supabase (PostgreSQL)	Feature Store	Serverless PostgreSQL with instant sub-80ms REST ingestion
Hopsworks	Model Registry	Cloud model artifact versioning, scalars & metadata tracking
AQICN + Open-Meteo	Data Ingestion	Live AQI telemetry & 72-hour weather forecast parameters
Streamlit + Flask	Web UI & API	Full-width glassmorphic dashboard + RESTful microservice
GitHub Actions	CI/CD Automation	Hourly feature updates & daily automated retraining
SHAP & Optuna	Explainability & AutoML	TreeExplainer feature attribution & automated parameter tuning

3. Architectural Rationale: Why Supabase for Feature Storage

While Hopsworks was initially evaluated for feature storage, critical operational roadblocks required adopting Supabase PostgreSQL as the primary serverless Feature Store:

- Free-Tier Quotas & Paywalls: Automated hourly ingestion (720 writes/month) quickly exhausts Hopsworks Cloud free credits, triggering HTTP 402 payment requirements or write suspensions. Supabase offers 500 MB of free storage, easily storing >5 years of hourly features with zero paywalls.
- Inactivity Hibernation: Free Hopsworks clusters hibernate and take 45-90s to wake up, causing 504 Gateway Timeouts in GitHub Actions. Supabase is always-on with sub-80ms response times.
- Dependency Footprint: Hopsworks HSFS client relies on heavy PySpark/JVM dependencies that cause memory issues on lightweight containers. Supabase communicates via lightweight PostgREST.
- Primary-Key Upsert: Supabase uses atomic ON CONFLICT (timestamp, city) DO UPDATE to prevent duplicate rows during cron retries.
- Division of Responsibilities: Hopsworks is retained for Model Registry (hsml), while Supabase handles high-frequency feature storage.

4. Modeling Strategy: Why XGBoost Over TensorFlow/LSTM

Deep learning sequence modeling via TensorFlow/Keras LSTM was evaluated, but XGBoost and Scikit-Learn ensembles were selected for production:

- Footprint: TensorFlow adds ~1.5 GB of binary wheels, crashing memory limits on free Streamlit Cloud and slowing CI/CD pipelines.
- Performance: On structured tabular time-series data with lag buffers (1h to 72h), XGBoost achieved superior R2 (0.8841)

and lower RMSE (28.45) compared to LSTMs.

3. Inference Speed: XGBoost computes 72-hour recursive multi-step predictions in under 15ms.

5. Model Performance Benchmarks

All models were trained using temporal cross-validation splits (80% train / 10% validation / 10% test) to prevent temporal data leakage.

Model Name	Algorithm	RMSE	MAE	R2 Score	Deployment Status
XGBoost	Gradient Boosted Trees	28.45	19.20	0.8841	Production Champion
Random Forest	Bagging Ensemble	31.80	21.65	0.8540	Primary Fallback
Ridge Regression	L2 Regularized Linear	42.15	29.80	0.7425	Statistical Baseline

6. Reliability, AutoML & SHAP Explainability

- Fallback Cascade: XGBoost -> Random Forest -> Ridge -> Persistence Baseline (guaranteeing 100% uptime).
- Optuna Hyperparameter Optimization: Automated tuning module searches tree depth, learning rate, subsample, and regularization parameters against validation RMSE.
- Top Predictive Drivers (SHAP): pm2_5_lag_1h (~38%), humidity_inversion (~19%), wind_speed_10m (~15%), is_smog_season (~14%), surface_pressure & traffic hours (~14%).

7. Verification & Deliverables Summary

- All project deliverables have been built, verified, and automated:
- [x] 100% Serverless Stack (GitHub Actions + Supabase + Hopsworks)
 - [x] 3-Day Recursive Forecast conditioned on future weather dispersion
 - [x] Live Streamlit Web App: <https://aqi-project-c3cg2qetkj3sdwcvzcdwcf.streamlit.app/>
 - [x] 72-Hour Interactive Forecast Drawer & CSV Download
 - [x] Optuna AutoML Hyperparameter Tuning Integration
 - [x] REST API with 7 endpoints (<http://localhost:5000>)
 - [x] 32/32 Automated Unit & Integration Tests Passing in PyTest